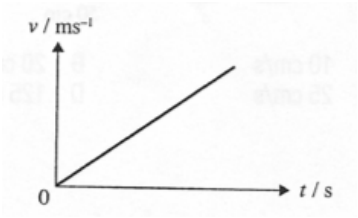
Questions for the tutor class:

2, 3, 4, 5, 7, 10, 11, 12, 14, 16, 19, 20, 23, 25, 26

1. A speed-time graph of an object is shown below



Which of the following best describe the graph?

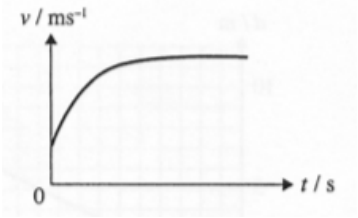
A. The object drops through vacuum

B. The object drops through air

C. The object is being thrown downwards through vacuum

D. The object is being thrown downwards through air

2. A speed-time graph of an object is shown below



Which of the following best describe the graph?

A. The object drops through vacuum

B. The object drops through air

C. The object is being thrown downwards through vacuum

D. The object is being thrown downwards through air

3. A stone is released from rest at the top of a 40 m high tower. Air resistance may be taken as negligible. What is the time taken for the stone to fall the last 10 m to the ground? (Take g as 10 m s^{-2}.)

A. 0.38 s

B. 1.4 s

C. 2.5 s

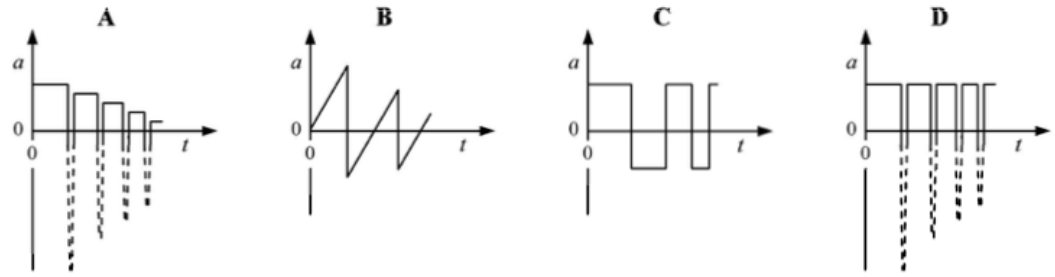
D. 2.8 s

4. A ball is thrown vertically upwards, rises to a maximum height, where it is momentarily at rest, and falls back to the hands of the thrower, The upward direction is to be taken as positive. Air resistance may be considered negligible. What is the acceleration of the ball at the various stages of its motion?

	rising	at maximum height	falling
A	$-9.81\,\mathrm{m\,s^{-2}}$	0	$+9.81\,\mathrm{m\,s^{-2}}$
B	$-9.81\,\mathrm{m\,s^{-2}}$	$-9.81\,\mathrm{m\,s^{-2}}$	$-9.81\,\mathrm{m\,s^{-2}}$
C	$+9.81\,\mathrm{m\,s^{-2}}$	$+9.81\,\mathrm{m\,s^{-2}}$	$+9.81\,\mathrm{m\,s^{-2}}$
D	$+9.81\,\mathrm{m\,s^{-2}}$	0	$-9.81\,\mathrm{m\,s^{-2}}$

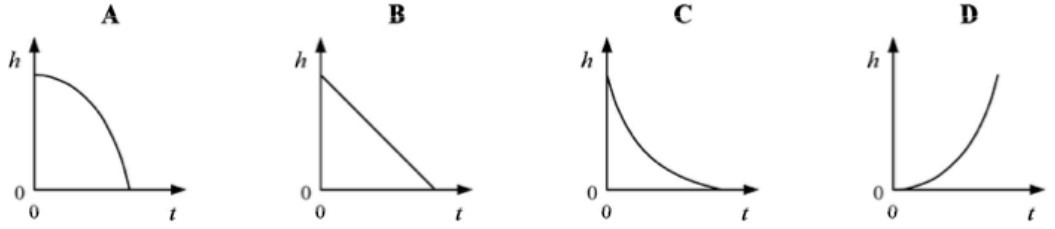
Solution: B

5. A steel ball is released from rest above a horizontal surface and bounces several times. Which graph best represents the variation of the ball's acceleration a with time t?



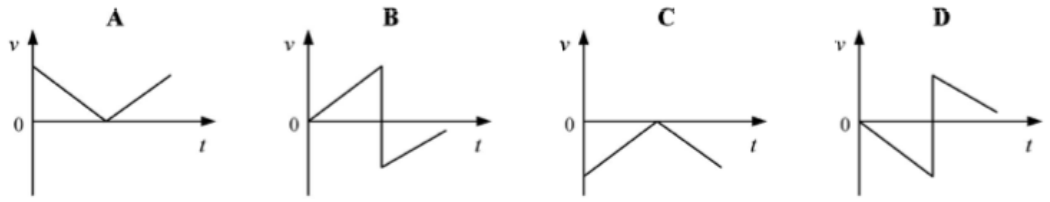
Solution: D

6. A small steel ball is released from rest and falls freely under gravity. If air resistance is negligible, which graph best represents the variation with time t of its height h above the ground?



Solution: A

7. A steel ball is released from rest so that it falls to the floor and bounces back. Which graph best represents the variation with time t of its velocity v, taking upwards as positive?



Solution: D

8. The acceleration of free fall of a body from rest is to be measured in an experiment. What are the measurements needed?

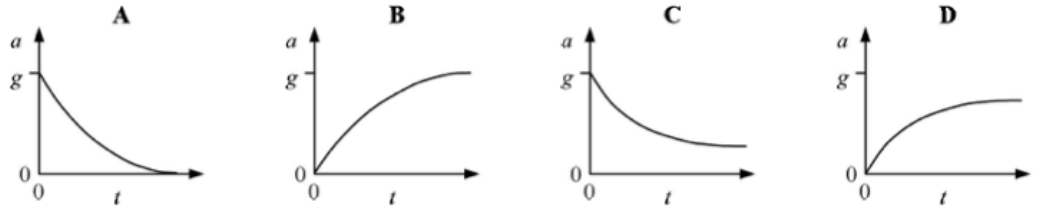
   A. the height of fall and the time of fall

   B. he height of fall and the weight of the body

   C. the mass of the body and the height of fall

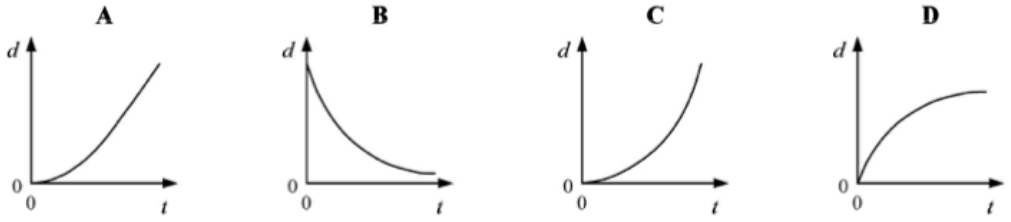
   D. the mass of the body and the time of fall

9. For a body falling from rest through air of uniform density, which of the following graphs shows how its acceleration a varies with time t?



Solution: A

10. A body falls from rest through the air. Which graph best illustrates the variation of the distance d fallen with time t?



Solution: A

11. A body is projected with an initial velocity u at an angle θ to the horizontal. At time t after projection, its height and horizontal distance from the point of projection O are denoted by y and x respectively. If air resistance may be taken as negligible, which pair of expressions give the values of y and x?

A $y = ut\cos\theta - \frac{1}{2}gt^2, \quad x = ut\sin\theta$
B $y = ut\sin\theta - \frac{1}{2}gt^2, \quad x = ut\cos\theta$
C $y = ut\sin\theta + \frac{1}{2}gt^2, \quad x = ut\cos\theta$
D $y = ut\cos\theta, \quad x = ut\sin\theta - \frac{1}{2}gt^2$

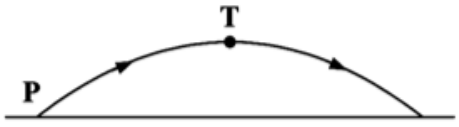
Solution: B

12. Two projectiles are launched from ground level with the same initial speed. The maximum height h_1 reached by projectile 1 is twice the maximum height h_2 reached by projectile 2. If θ_1 and θ_2 denote the respective launch angles, as measured from the horizontal, these angles satisfy which of the following relationships?

A. $\cos\theta_1 = \sqrt{2}\cos\theta_2$
B. $\sin\theta_1 = \sqrt{2}\sin\theta_2$
C. $\tan\theta_2 = \sqrt{2}\tan\theta_2$
D. $\cos\theta_1 = 2\cos\theta_2$

Solution: Answer is B

13. A ball, projected from P, follows a parabolic path and the highest point reached is T. Air resistance may be assumed to be negligible. Which statement is applicable to the vertical component of acceleration of the ball?



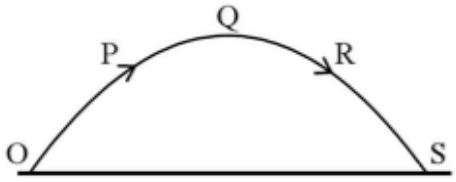
A. It is zero at T

B. It is greatest at T

C. It is greatest at P

D. It is the same at P as at T

14. The given diagram illustrates a projectile launched at point O following the path OPQRS. Air resistance may be neglected. When the projectiles is at the highest point Q of its path, which of the following statements is true?



A. The horizontal component of the projectile's acceleration is zero

B. The horizontal component of the projectile's velocity is zero

C. The kinetic energy of the projectile is zero

D. The momentum of projectile is zero

15. The diagram shows a projectile fired at speed u at an angle a to the horizontal. The acceleration of free fall may be taken as g. Which line in the table correctly shows the vertical and horizontal components of its velocity after a time t ?



	vertical component	horizontal component
A	$u\sin\alpha$	$u\cos\alpha$
B	$u\sin\alpha - gt$	$u\cos\alpha - gt$
C	$u\sin\alpha - gt$	$u\cos\alpha$
D	$u\cos\alpha$	$u\sin\alpha - gt$

Solution: $v_{vertical}$ = (u sina) + (-g)t = u sins - gt,
$v_{horizontal}$ = (u cosa) + (0)t = u cosa

16. An object drops from the top of a building and takes 5.0 s to reach the ground. The acceleration of free fall is 10 m/s^2.

(a) What is the speed of the object after 1.0 s; 2.0 s and 5.0 s?

(b) What is the distance travelled after:

- in the first second
- in the next second?

(c) What is the height of the building?

Solution:

(a) (i) 10.0 m/s : (ii) 20.0 m/s; (iii) and 50.0 m/s

(b) (i) 5.0 m ; (ii) 20.0 m

(c) 125 m

17. A bullet is fired vertically upwards from a gun with an initial speed of 100 m/s. The acceleration of free fall is 10 m/s^2.

(a) What is the speed of the bullet after 1.0 s?

(b) What is the time taken for the speed to decrease to zero?

(c) What is the speed of the bullet when it reaches the highest point?

(d) What is the average speed of the bullet for the whole journey upwards?

(e) what is the maximum height reached by the bullet?

Solution:

(a) 90 m/s

(b) 10 s

(c) speed = 0 at the highest point

(d) 50 m/s

(e) 500 m

18.

(a) Without opening his parachute, a sky diver jumps from a plane

- Describe the motion of the sky diver as he falls through the air.
- Explain why he has to spread himself wide.

(b) Explain why he is able to safely after the parachute opened ?

Solution:

(a) As the sky driver falls through the air:

- Describe: His velocity increases $\rightarrow$ The air resistance increase $\rightarrow$ his acceleration decreases. When the acceleration is zero, he falls at his uniform terminate velocity.
- He spreads himself wide to increase cross-sectional area $\rightarrow$ the air resistance increases $\rightarrow$ the terminal velocity is smaller.

(b) With the parachute opened, there is a large increase in the cross-sectional area. The air resistance becomes very large and much lower terminal velocity is achieved $\rightarrow$ safe !

19. The variation of the velocity v of an object falling in air with time t is shown in the table

$t(s)$	0	1.0	2.0	3.0	4.0	5.0	6.0
v m/s	0	9.5	15.0	19.0	21.5	23.0	23.0

(a) Calculate the average acceleration of the object

- from t = 0 to t = 1.0 s
- from t = 1.0 s to t= 2.0 s

(b) Explain the difference between the acceleration in (a) above for two cases.

(c) What is the terminal velocity?

Sketch an acceleration-time graph for the motion of the object.

Solution:

(a) (i) 9.5 m/s^2; (ii) 5.5 m/s^2

(b) Velocity increases $\rightarrow$ air resistance increases $\rightarrow$ average acceleration decreases

(c) 23.0 m/s

20. In a speed trap, two pressure-activated strips are placed 120 m apart on a highway on which the speed limit is 85 km/h. A driver going 110 km/h notices a police car just as he/she activates the first strip, and immediately slows down. What deceleration is needed so that the car's average speed is within the speed limit when the car crosses the second strip?

Solution:

$$a = -2.73\ m/s^2$$

21. A shell is fired from the ground with an initial speed of $1.70 \times 10^3\ (m/s)$ (approximately five times the speed of sound) at an initial angle of 55.0° to the horizontal. Neglecting air resistance, find
    (a) the shell horizontal range
    (b) the amount of time the shell is in motion.

Solution:

(a) $R = 2.77 \times 10^5\ m$

(b) $t = 284\ s$

22. A daredevil is shot out of a cannon at 45.0° to the horizontal with an initial speed of $25.0\ (m/s)$. A net is positioned at a horizontal distance of $50.0\ (m)$ from the cannon from which it is shot. At what height above the cannon should the net be placed in order to catch the daredevil?

Solution:

$$h = 10.76\ m$$

23. Legend has it that Gallileo tested out his newly developed theory of projectile motion by throwing weights from the top of the Leaning Tower of Pisa. (No wonder he eventually got into trouble with the authorities!) Suppose that, one day, Gallileo simultaneously threw two equal weights off the tower from a height of $100\ m$ above the ground. Suppose, further, that he dropped the first weight straight down, whereas he threw the second weight horizontally with a velocity of $5\ m/s$. Neglect the effect of air resistance, find:
    (a) Which weight struck the ground first?
    (b) How long, after it was thrown, did it take to do this?
    (c) What horizontal distance was traveled by the second weight before it hit the ground?

Solution:

(a) Both weights strike the ground simultaneously.

(b) $t = 4.515\ s$

(c) $R = 22.58\ m$

24. A cannon placed on a 50 m high cliff fires a cannonball over the edge of the cliff at $v = 200\ m/s$ making an angle of $\theta = 30^\circ$ to the horizontal. How long is the cannonball in the air? Neglect air resistance.

Solution:

$$t = 20.88\ s$$

25. A car travel due East with a speed of 50 km/h. Rain is falling vertically with respect to Earth. The traces of the rain on the side windows of the car make an angle of 60.0° with the vertical. Find the velocity (including the magnitude and the direction) of the rain with respect to the following:
    (a) the car,
    (b) Earth.

Solution:

(a) $57.7\ km/h$

(b) $28.9\ km/h$

26. An escalator is 20.0 m long. If a person stands on the escalator, it takes 50.0 s to ride from the bottom to the top.
    (a) If a person walks up the moving escalator with a speed of $0.500\ m/s$ relative to the escalator, how long does it take the person to get to the top?
    (b) If a person walks down the "up" escalator with the same relative speed as in item (a). How long does it take to reach the bottom?

Solution:

(a) $t_1 = 22.2\ s$

(b) $t_2 = 200\ s$